Validating a synthetic segmentation benchmark by rank correlation: how the methods compared, the clinical reference and training noise change the answer in wound imaging

Abstract

Synthetic test sets for medical image segmentation are often validated by showing that they rank methods like a clinical dataset. We measured how this correlation depends on the methods compared, the clinical reference and training noise. We scored 65 methods on 11 public clinical datasets in four modalities and on synthetic wound images: 18 untrained classical methods, 14 methods trained from scratch on wound photographs, the same 14 trained on colonoscopy, and 19 modern models (18 ImageNet-pretrained architectures with up to three training seeds, and nnU-Net). Agreement was the Spearman correlation of per-method mean Dice. Pooling classical and modern methods made four wound datasets agree at 0.82, although they agreed at 0.35 among the classical and 0.21 among the modern methods. A synthetic benchmark agreed with the wound datasets at 0.65 over the pooled methods but at 0.21 among the modern models, where it could not be shown to be better or worse than another wound dataset. Two training seeds of the same architectures agreed at only 0.44 on the same wound dataset. The model ranked first on the test set from the training clinic lost 0.101 Dice on average on other wound datasets, against 0.075 for a random choice. On the AZH dataset the pretrained models, run on images resized to 256 pixels, missed 28.7% of wounds on average, against 1.5% for nnU-Net at native resolution. Validity should be reported within the group of methods to be compared, against several clinical datasets.

Keywords Benchmark validity; Rank agreement; Synthetic data; Wound imaging; Image segmentation; Training variance

Introduction

Segmentation software is increasingly evaluated before it reaches a clinic, by vendors comparing candidate models, by hospitals validating a product on local data and by regulators considering computational evidence. Synthetic and simulated images are attractive for this purpose because they come with exact ground truth and can be generated in any quantity, and in-silico trials and edited images have already been used to evaluate and stress-test medical models [1–4]. Rankings of segmentation methods are, however, known to be fragile. They change with the test cases, the metric, the aggregation rule and the annotators [5–7], with the choice of benchmark [8, 9] and with the hospital the test data come from [10, 11].

The usual evidence that a synthetic test set is trustworthy is a rank correlation between method scores on the synthetic set and on one clinical dataset, sometimes expressed relative to the agreement between two real datasets [12, 13]. Synthetic data have been proposed for model validation and selection on a similar basis [14–16], and in computed tomography (CT) the transfer of rankings from synthetic to real data has been shown to depend on the task [17]. Such a correlation is computed over a particular set of methods, which we call the method pool, and we call a benchmark's rank agreement with clinical data its measured validity. Outside medical imaging it is established that agreement between benchmarks depends on which models are compared [18], and that the correlation between in-distribution and out-of-distribution performance depends on how the models were selected and trained [19–21]. When groups of methods

change order between domains, pooling them produces correlations that hold in neither group, a form of Simpson's paradox [22, 23].

Medical segmentation studies often pool exactly such groups. Wound segmentation work, for example, spans classical colour and threshold methods and trained networks [24], and recent wound studies compare several deep architectures across datasets [25, 26]. A synthetic benchmark validated over a pool that mixes such groups inherits whatever the pooling does to the correlation. A second source of variation lies in training itself. Retraining the same network with another random seed changes its score, and for closely matched models these changes can be as large as the differences between models [27, 28]. A ranking of modern models on one dataset may then be partly a ranking of training runs, and training noise limits how closely two datasets can be expected to agree on an order. Decision thresholds add a further choice, because networks differ in how confident their outputs are [29]. How large these effects are for segmentation, and what they imply for validating a synthetic benchmark, has not been measured.

We scored 65 methods in four groups, each with one training regime, on 11 clinical datasets from eight sources in four modalities and on synthetic wound benchmarks from three generator families. The modern group contains 18 ImageNet-pretrained architectures, each trained with up to three random seeds, and nnU-Net. We asked how much pooling groups of methods changes agreement between clinical datasets, whether agreement is organised by modality within groups, how much of the disagreement among modern models is training noise, and what this implies for the measured validity of a synthetic benchmark and for choosing a model. The contributions are:

1. A measurement of the composition effect in segmentation benchmark agreement. Pooling untrained with modern methods raises agreement between wound datasets from 0.35 and 0.21 within the groups to 0.82, and an exact identity shows that ranking methods within their groups before pooling returns the size-weighted average of the within-group agreements.

2. A demonstration that the measured validity of a synthetic benchmark follows the method pool and the clinical reference: the same benchmark agrees with wound data at 0.65 or at 0.21 depending on whether untrained methods are in the pool, and its agreement with single wound datasets varies within one pool.

3. Evidence from 18 architectures with up to three training seeds that rankings of modern models on wound data are dominated by small differences and training noise, with the consequences for model selection, including selection on a test set from the training clinic, and a failure mode on one dataset that mean Dice hides.

Materials and Methods

Clinical datasets

Eleven public datasets serve as clinical references (Table 1). The wound photographs are Medetec [30], CO2Wounds-V2 [31], the validation set of the Foot Ulcer Segmentation (FUSeg) challenge [32] and the AZH chronic wound dataset [33]. FUSeg and AZH were collected at one wound centre and are treated as one source; CO2Wounds-V2 contains 764 smartphone photographs from 96 leprosy patients in Colombia [31] and shares no source with any other dataset. The colonoscopy datasets are Kvasir-SEG [34] and its sessile subset, CVC-ClinicDB [35], CVC-ColonDB [36] and ETIS-LaribPolypDB [37]; the two CVC sets come from one group and are one source. PH2 [38] is the dermoscopy reference and the TN3K test split [39] the thyroid ultrasound reference. A disjoint slice of the AZH training set serves as a within-dataset control.

The FUSeg reference is the first 120 validation images and the AZH reference the first 150 test images, sorted by file name; Medetec is used in full. The Kvasir reference was drawn only from images outside the sessile subset, so the two Kvasir references share no image. For CO2Wounds-V2 and the five other non-wound datasets, 150 image and mask pairs were taken evenly spaced over the files sorted by number, because consecutive colonoscopy frames are near duplicates. Images in these six sets were resampled so that the long side is at most 256 pixels; wound masks were binarised at any non-zero value and all others above 127. Several images can come from one photograph: Medetec's 152 images are 38 photographs with four variants each, and the 146 AZH test images and 78 control images are crops of 118 and 71 photographs. The photograph is therefore the resampling unit for these three sets in every analysis, and the image for all others.

Duplicated and overlapping evaluation images were removed before any analysis. In FUSeg, 8 validation images are byte-identical to FUSeg training images (7) or to another validation image (1). FUSeg and AZH come from the same clinic, so every AZH test image and every control image was compared with all 1,010 FUSeg photographs (810 training, 200 validation) by feature matching: scale-invariant feature transform (SIFT) keypoints, a ratio test at 0.75, and a similarity transform fitted by random sample consensus (RANSAC), with at least 25 geometrically consistent keypoints counting as a match. The rule for removal was the same for every set: an evaluation image was removed if it is a copy or a crop of a photograph used elsewhere in this study, that is a FUSeg training image used for training or threshold tuning, or an image of another evaluation set. Of the 150 AZH test images, 61 are crops of FUSeg photographs; 4 of these were cut from training images and were removed, and the other 57 come from FUSeg photographs that no model saw and that are not among the FUSeg reference images, so they were kept, leaving 146. Of the 150 control images, 70 are crops of FUSeg photographs used in this study (51 from its training split and 19 from the validation images used as reference), and 2 more are pixel-identical to AZH test images; they were removed, leaving 78. One further control image is a crop of a FUSeg photograph used nowhere in the study (a duplicate removed from both the training and the reference images) and was kept. A fifth wound dataset (WoundSeg) was prepared but excluded before any method was scored on it, because 322 of its 2,686 masks did not match the size of their image, so the pairing of images and masks could not be trusted. The exclusions and the overlap check are listed in Online Resource 1 (Tables S9a and S9e).

Table 1 Clinical datasets after removal of duplicated and overlapping images. Units are the resampling units (photographs for Medetec and the AZH sets, images otherwise). Lesion fraction is the median share of image pixels inside the reference mask

Dataset	Modality	Source	Images	Units	Empty masks	Lesion fraction
Medetec	Wound photography	Medetec archive	152	38	0	0.098
CO2Wounds-V2	Wound photography	Leprosy clinic, Colombia	150	150	0	0.045
FUSeg (validation)	Wound photography	AZH Wound Center	112	112	4	0.007
AZH (test)	Wound photography	AZH Wound Center	146	118	10	0.018
AZH train slice (control)	Wound photography	AZH Wound Center	78	71	6	0.009
Kvasir	Colonoscopy	Kvasir collection	120	120	0	0.128
Kvasir-sessile	Colonoscopy	Kvasir collection	120	120	0	0.043
CVC-ClinicDB	Colonoscopy	CVC, Barcelona	150	150	0	0.067
CVC-ColonDB	Colonoscopy	CVC, Barcelona	150	150	0	0.038
ETIS-Larib	Colonoscopy	ETIS, Paris	150	150	0	0.020
PH2	Dermoscopy	Porto	150	150	0	0.231
TN3K (test)	Thyroid ultrasound	TN3K	150	150	0	0.066

Method groups

The 65 methods form four groups that differ in training and in training domain (Table 2; every method is listed in Online Resource 1, Table S1). The 18 untrained methods (U) are classical thresholding, colour, clustering and region-growing algorithms with no learned parameters. The 14 from-scratch methods (F) are three pixel classifiers (two logistic regressions and gradient boosting on colour and texture features), a three-level U-Net with 117,681 parameters, and ten architectures from segmentation_models_pytorch [40] with randomly initialised encoders. These ten are U-Net [41], U-Net++ [42], DeepLabV3+ [43], FPN [44], PSPNet [45], LinkNet [46] and MA-Net [47] on ResNet-18 [48], U-Net on MobileNetV2 [49] and EfficientNet-B0 [50], and SegFormer on MiT-B0 [51].

The from-scratch group was trained on FUSeg training images. The pixel classifiers were fitted on 247,975 pixels sampled from images 0 to 249, balanced between lesion and background. The threshold of one logistic regression was tuned on images 250 to 399 (best value 0.90), and this value was used for all three pixel classifiers. The small U-Net uses a threshold of 0.5. The ten architectures were trained on images 0 to 392 at 128 by 128 pixels for 30 epochs (Dice plus binary cross-entropy, Adam, learning rate 0.001, horizontal flips), and each threshold was tuned on images 400 to 549. The Kvasir-trained group (K) repeats this protocol exactly on Kvasir-SEG images that appear in neither Kvasir reference nor the sessile subset (pixel threshold 0.60).

The modern group (M19) contains 18 architectures with ImageNet-pretrained encoders [52] and nnU-Net. Ten are the architectures of the from-scratch group with pretrained encoders. Eight were added to cover larger encoders and current designs: U-Net, DeepLabV3+ and MA-Net on ResNet-50, U-Net on EfficientNet-B3 [50] and on Swin-T [53], SegFormer on MiT-B2 [51], UPerNet [54] on ConvNeXt-T [55] and FPN on RegNetY-1.6GF [56]. All 18 were trained with one protocol on FUSeg training images 0

to 392 with the one image duplicated in the evaluation set removed (392 of the 810 FUSeg training images). Images were letterboxed so that the long side is 256 pixels, with the aspect ratio kept and the remainder padded with black pixels. Training used Dice plus binary cross-entropy, AdamW (learning rate 0.0003, weight decay 0.0001, cosine decay), batches of 16, 40 epochs, random horizontal and vertical flips and rotations by multiples of a right angle, and mixed-precision arithmetic with gradient scaling. Each model's decision threshold was chosen on a grid from 0.05 to 0.90 in steps of 0.05 to maximise mean Dice on 148 held-out FUSeg training images (images 400 to 549 without two duplicates of evaluation images); this held-out Dice is optimistic, because the threshold was tuned on the same images. Held-out Dice of the 51 individual models ranged from 0.75 to 0.84.

Each architecture was trained with three random seeds (0, 1 and 2). The seed sets the initial weights of the parts not taken from ImageNet, the order of the training batches and the augmentation draws. Seed 0 of the ten original architectures was trained in a first session on a graphics processing unit (GPU); seeds 1 and 2 of these and all seeds of the eight new architectures were trained in a second session with the same training code. The second session recorded segmentation_models_pytorch 0.5.0, timm 1.0.22 and PyTorch 2.11.0; the first did not record the PyTorch and timm versions. Of the 44 trainings in the second session, 41 produced a usable model. The 3 that did not were MA-Net trainings (3 of 5: ResNet-18 at seed 2 and ResNet-50 at seeds 0 and 2), in which mixed-precision arithmetic produced non-finite values; the other 39 trainings had none. Failed trainings were not imputed. In the main analyses each architecture is represented by the per-image mean of its scores over the seeds that exist (its seed average), so MA-Net on ResNet-18 averages two models and MA-Net on ResNet-50 is a single model.

nnU-Net v2 [57] was trained once, in its 2D configuration with the default trainer, on all 392 training images (one model, no cross-validation folds) for 100 epochs instead of the default 1,000. Its foreground probability is thresholded at 0.5; its Dice on the 148 held-out images is 0.83. To produce inputs for the robustness analyses nnU-Net was run again in the second session on the same images. Per-image Dice differed from the first run by at most 0.0034 and dataset means by at most 0.00003, with no change of rank, so all main analyses use the first-run scores (Online Resource 1, Table S9b).

The Segment Anything Model (SAM) [58] and its medical version MedSAM [59] were also run, with box prompts taken from the reference masks. Because those prompts give them the lesion location, they are reported only in Online Resource 1 (Table S8) and are not part of any method set.

In the main post-processing pipeline (P0), untrained methods receive morphological closing and opening with a 5-pixel elliptical element. Methods that output a probability map receive Gaussian smoothing with a standard deviation of 2 pixels before thresholding. Every method then has connected components smaller than 0.05% of the image removed.

Table 2 Method groups. Mean Dice is the range of group means (mean over methods of the per-method mean) over the four wound datasets and over the seven other clinical datasets

Group	n	Training data	Training seeds	Mean Dice, wound	Mean Dice, other
Untrained (U)	18	none	none	0.06 to 0.30	0.07 to 0.49
From scratch (F)	14	FUSeg	one	0.28 to 0.53	0.01 to 0.13
Kvasir-trained (K)	14	Kvasir-SEG	one	0.08 to 0.29	0.07 to 0.64
Modern (M19)	19	FUSeg, ImageNet encoders	up to three per architecture; one for nnU-Net	0.44 to 0.80	0.01 to 0.07

Synthetic benchmarks

Three parametric generator families produce wound images with exact masks, 45 images per benchmark in three severity classes. Generator A places a lesion on a uniform drape and reproduces the benchmark released with an earlier system description [60]. Generator B renders textured skin with clutter such as hair and gauze, and its background texture amplitude is a parameter; it was built at amplitudes 11, 28 and 40. Generator C shares no code with B and uses spectral noise, vignetting and a colour mixture model. Every benchmark was built at three synthetic seeds (2024, 7 and 101). Background texture is the median local standard deviation of shading-corrected luminance in a 9-pixel window over background pixels, measured at 512 pixels.

Agreement between image sets

Each method is scored per image with the Dice similarity coefficient (Dice) between its mask and the reference mask. Agreement between two image sets is the Spearman correlation between their vectors of per-method mean Dice over a given set of methods: a rank correlation, so 1 means that the two image sets order the methods identically and 0 means no relation. The 55 pairs of the 11 clinical datasets are grouped by level: within source (one clinic or group, no shared images; 3 pairs), within modality from different sources (13 pairs) and across modality (39 pairs). The within-dataset control compares AZH with the disjoint AZH slice. The modality difference is mean same-modality agreement (16 pairs) minus mean across-modality agreement (39 pairs). For the composition analyses each pair is assigned to one of four categories relative to a domain, the modality a trained group was trained on. For wound photography these are domain with domain (the 6 wound pairs, whose mean we call wound-pair agreement), domain with other datasets (28 pairs), other datasets of the same modality (the 10 colonoscopy pairs) and other datasets across modality (11 pairs). Method sets are the four groups alone and the pooled sets U+F (32 methods), U+K (32) and U+M19 (37). Every analysis was repeated with normalised surface Dice (NSD), the share of the predicted and reference boundaries that lie within a tolerance of each other, at 2 pixels [61], and with the images whose reference mask is empty excluded.

Pairs share datasets, so tests on pairs overstate the evidence. The modality difference was therefore tested by assigning the modality labels to the 11 datasets in every distinct way (13,860 assignments) and counting how often the difference was at least as large as observed. Swapping the labels of the two single-dataset modalities gives the same partition of pairs, so the smallest attainable p is 2/13,860 = 0.00014. Because the three within-source pairs could carry the result alone, the test was repeated with them dropped under every relabelling (52 pairs). The test treats the method set and the images as fixed. For pooled sets it shows that pooling produces apparent modality structure, not evidence that such structure exists.

Two bootstrap designs give 95% percentile intervals. The method bootstrap resamples methods with replacement within each group, so a pooled set keeps its composition (2,000 draws). These intervals show how much agreement depends on which methods of each kind are included, treating the methods as a sample of comparable methods. Spearman correlations use average ranks for ties, which occur when a method is drawn more than once; with 7 to 19 methods a draw contains few distinct methods, so these intervals are approximate. The hierarchical bootstrap also resamples the units of every dataset independently (1,000 draws for the clinical and seed analyses; 2,000 for the synthetic-benchmark analyses, which also resample the synthetic images), using photographs as units for Medetec (38 photographs), AZH (118) and the control (71), and images for the other datasets. Resampling units adds noise to every method mean and so pulls a single agreement value towards zero in each draw. Hierarchical intervals are therefore conservative for claims that a single agreement value exceeds zero,

but not for claims that agreement is low, and not for differences in which one term shares images across seeds (see the training-seed analyses). Intervals in the text are method-bootstrap intervals unless marked hierarchical; miss-rate intervals resample units only, and seed-analysis intervals resample architectures. Analyses of the synthetic benchmarks that involve untrained or from-scratch methods use the method bootstrap only, because per-image synthetic scores exist only for the GPU-trained groups. Image sampling within a dataset was also summarised by split-half reliability, an estimate of how well a new set of images from the same dataset would rank the methods again: the Spearman correlation between method means on two random halves of the units, corrected with the Spearman-Brown formula (median of 500 splits).

Stratified agreement and why it equals the within-group average

Stratified agreement ranks methods within their own group first and then pools them. Within each group g of size n_g, a method's mean on a dataset is replaced by its within-group percentile $(r - 0.5)/n_g$, where r is its rank in the group, and agreement is the Pearson correlation of these percentiles over the pooled set of $N = \Sigma\, n_g$ methods. The percentiles have mean 1/2 in every group, so the pooled mean is also 1/2 and the between-group part of the pooled covariance is zero. The pooled covariance of the percentiles on datasets A and B is therefore $(1/N)\, \Sigma\, n_g\, c_g$, and each pooled variance is $(1/N)\, \Sigma\, n_g\, v_g$, where c_g and v_g are the covariance and variance within group g. Without ties, $v_g = (1 - 1/n_g^2)/12$ on every dataset, and within a group the correlation of percentiles is the Spearman correlation r_g. Hence

$$\text{stratified agreement} = \Sigma_g\, w_g\, r_g \,/\, \Sigma_g\, w_g, \quad \text{with} \quad w_g = n_g - 1/n_g,$$

a size-weighted average of the within-group agreements: the pooled within-group correlation, applied to within-group ranks. The weights do not depend on the pair, so the identity also holds for means over pairs. It holds exactly when no two methods in a group have the same mean on either dataset. With ties, each group is weighted by n_g times the square root of its two percentile variances, and the weights then depend on the pair. In our data ties occur only on TN3K and PH2; there the Pearson value differs from the weighted average by at most 0.006 for a single pair and by at most 0.001 for a mean over pairs, and for the wound pairs the identity is exact. Stratified agreement thus adds no information beyond the within-group agreements; what it shows is how much of a pooled value comes from differences in level between groups. Ranking the pooled percentiles again (a Spearman version) changed single pairs by at most 0.03 under Dice.

Training-seed analyses

In total 16 architectures have all three seeds: nine of the original ten and seven of the eight new ones. The two MA-Net architectures lack seeds and nnU-Net has one training, so these three are left out of the seed analyses. For each wound dataset, same-dataset cross-seed agreement is the mean Spearman correlation between the architecture means of seeds s and t on the same images, over the three seed pairs. Its counterpart across datasets correlates seed s on one wound dataset with a different seed t on another (36 correlations over the 6 wound pairs), so that training noise is present in both and only the dataset differs. The gap between the two measures how much more two datasets disagree than two trainings scored on the same images do. Because the two seeds share their test images, the gap also contains image-sampling noise and is an upper bound on the part of the disagreement due to the datasets; and because both seeds share one resample of units in each hierarchical draw, hierarchical intervals for the gap are shifted upwards rather than towards zero. We also compared wound-pair agreement at single seeds (mean over the three seeds) with agreement of the seed averages, and set the spread of architecture means on each dataset against the seed-to-seed standard deviation of one architecture's mean (pooled over the 16

architectures). The reliability of an architecture's mean Dice was summarised by a one-way random-effects intraclass correlation (ICC) over the 16 architectures: with W the mean seed variance and T the variance of the three-seed means minus W/3, the ICC is T/(T + W) for one seed and T divided by the variance of the three-seed means for a three-seed average. Intervals resample architectures (2,000 draws); hierarchical intervals also resample units (1,000 draws).

One reading rule was written down after the new models had been trained and scored, but before any seed analysis was run (the pre-specified rule). The statement that modern models disagree across wound datasets because the datasets differ, not because of training noise, would be supported if same-dataset cross-seed agreement exceeded 0.7 while cross-dataset agreement stayed near 0.1 to 0.3. If cross-seed agreement were as low as cross-dataset agreement, the modern models would be treated as not distinguishable. The seed analyses are detailed in Online Resource 1 (Table S4).

Synthetic benchmark analyses

For each wound reference, benchmark agreement is the Spearman correlation between generator B at amplitude 40 and that reference, averaged over the three synthetic seeds. It is compared with the mean agreement of the other three wound datasets with the same reference; the difference (benchmark minus other) is positive when the benchmark ranks methods more like the reference than another wound dataset does. Averaged over the four references, the second term equals wound-pair agreement. The texture effect is benchmark agreement at amplitude 40 minus agreement at amplitude 11. Benchmark agreement was also computed for single-seed versions of the original architectures, of the new architectures and of the full modern group, in which every architecture is represented by its model at one training seed; only the full single-seed modern group also includes nnU-Net.

Model selection and failure modes

For the modern models, the selection loss from dataset A to dataset B is the best mean Dice on B minus the mean Dice on B of the candidate ranked first on A. The expected loss of a random choice is the mean over candidates of the best mean Dice minus the candidate's. The candidates are the seed-averaged architectures and nnU-Net, and, because a seed average is not the output of any model, also the single-seed models of each training seed. The synthetic chooser picks the candidate with the highest mean Dice on generator B (mean over synthetic seeds). Selection losses are small Dice differences and are given to three decimals; their intervals resample the units of each dataset with the candidates fixed (2,000 draws). A miss is a non-empty reference mask segmented with Dice below 0.1. Miss rates were computed for every individual trained model and averaged over seeds per architecture; their intervals resample units (2,000 draws) with the models fixed. To look for the cause of the misses on AZH we related, for each image, the share of pretrained models that miss it to its share of black padding pixels (every colour channel at most 8) and to its lesion fraction, and we measured from the feature matches the scale of AZH crops relative to the FUSeg photographs they were cut from.

Robustness analyses

The main analyses were repeated with two other post-processing pipelines (P1: threshold, closing and opening with a 5-pixel element and small-component removal; P2: threshold and small-component removal only), with NSD at tolerances of 1, 3 and 5 pixels and at 1% of the image diagonal, and with FUSeg resampled so that its long side is 256 pixels, the resolution of most other datasets. For the modern models the threshold protocol was also varied: one common threshold for all models (each value of the grid from 0.05 to 0.90), and a per-dataset oracle threshold for every model, chosen on the evaluation

images themselves to reduce calibration differences, that is differences between models in the level of their output probabilities. These reanalyses used probability maps rescored in the second session; for the pooled sets the untrained and from-scratch methods were rescored locally with the same variants. Three Kvasir-trained pixel classifiers were not rescored, so robustness results for the Kvasir group use the remaining 11 methods (Online Resource 1, Table S6).

Primary and exploratory analyses

Two analyses were planned as primary. The composition effect was assessed with the paired method-bootstrap interval of pooled agreement minus the higher within-group agreement, and the modality structure within groups with the exact relabelling test. The modern group was redefined from 11 to 19 methods when the additional trainings became available, before the analyses reported here were run. With 7 method sets tested, every modality difference in Table 3 remains significant at the 5% level after Holm correction (largest adjusted p 0.0092). All other analyses, including the training-seed, benchmark, selection and robustness analyses, are exploratory; they are reported with intervals and are not corrected for multiple comparisons.

Large language model assistants were used to write analysis code and to draft and edit the text; all code, numbers and text were checked by the authors.

Results

The groups differ in level and in domain

The groups differ in level and in domain (Table 2). The modern models reach group mean Dice of 0.44 to 0.80 on the wound datasets but 0.01 to 0.07 on the seven other datasets, and the from-scratch group shows the same pattern at a lower level. The Kvasir-trained group leads on colonoscopy (0.21 to 0.64) and is close to the untrained group on wound photographs (0.08 to 0.29 against 0.06 to 0.30). We call a group mean Dice below 0.1 on a dataset the floor, because rankings there order near-zero scores. With the trained models fixed, each dataset ranks the methods reproducibly across its images: split-half reliability within the untrained, from-scratch and Kvasir-trained groups is 0.84 to 0.99 on the 11 datasets, and 0.82 to 0.92 for the modern models on the four wound datasets. This reliability does not include retraining.

Pooling creates agreement and disagreement

Pooling the untrained group with the modern group changes agreement in both directions (Fig. 1a, Table 3). Over the pooled 37 methods the four wound datasets agree at 0.82 (0.77 to 0.87), against 0.35 (0.12 to 0.59) among untrained methods and 0.21 (−0.07 to 0.49) among modern models; the pooled value exceeds the higher of the two by 0.47 (0.26 to 0.62). Agreement between wound and other datasets becomes −0.56 (−0.66 to −0.45), against 0.09 and 0.14 within the groups, and non-wound datasets of different modalities appear to agree at 0.43 (0.23 to 0.62), against 0.04 and 0.01. Under NSD the pooled wound-pair agreement is 0.88, against 0.62 and 0.48 within the groups. The full agreement matrices are shown in Online Resource 1 (Fig. S1).

The from-scratch group moves agreement in the same direction by less (Fig. 1b). Pooled with the untrained methods, wound datasets agree at 0.52, against 0.35 and 0.16 within the groups, but the excess over the untrained group, 0.17, has an interval that includes zero (−0.06 to 0.35). Agreement between wound and other datasets turns negative, −0.26 (−0.44 to −0.09), against 0.09 and 0.14 within the groups.

The mechanism is visible in the group means. The modern models lead the untrained methods on every wound dataset and fall below them on every other dataset (Table 2; per-dataset means in Online Resource 2), so a pooled correlation between a wound and a non-wound dataset is dominated by which group each method belongs to, without any pair of methods changing order within a group. The Kvasir-trained group is the mirror experiment. It leads on colonoscopy, but on wound photographs its means (0.08 to 0.29) are close to those of the untrained methods (0.06 to 0.30), so the two groups do not change places across modalities. Pooling it with the untrained group pulls agreement between colonoscopy and other datasets below both groups (0.02, against 0.13 and 0.20) but does not reverse its sign (Fig. 1c).

Stratified agreement removes the effect and, as derived in the Methods, equals the size-weighted average of the within-group agreements. For U+M19 the stratified wound-pair agreement is 0.28 (size-weighted within-group average 0.28; pooled 0.82), and the stratified modality difference is 0.41, against 1.10 pooled. For U+F and U+K the stratified modality differences are 0.54 and 0.55, against 0.83 and 0.68 pooled.

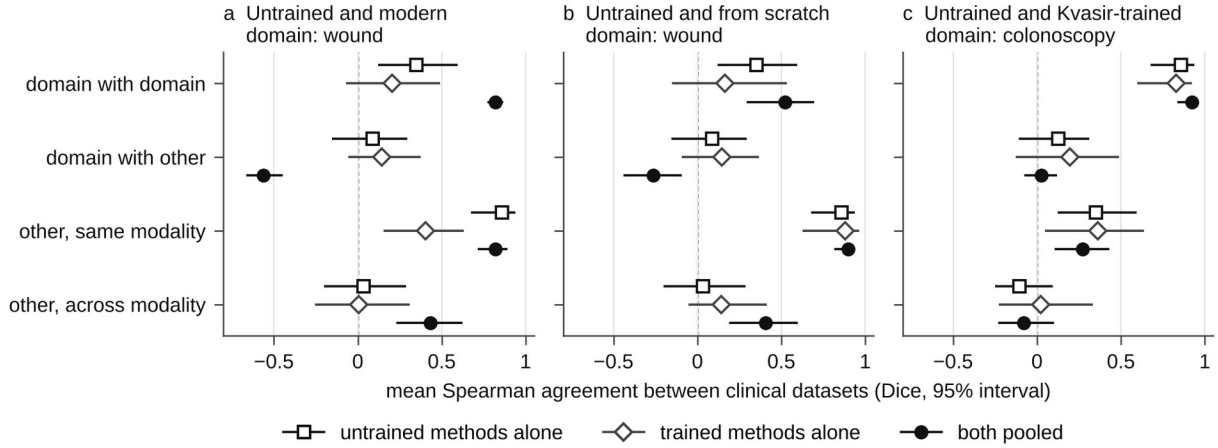


Fig. 1 Mean Spearman agreement between pairs of clinical datasets by category, for each group alone and pooled (Dice; 95% method-bootstrap intervals). The domain is the modality the trained group was trained on. a: untrained (U) and modern (M19) methods. b: untrained and from-scratch (F) methods. c: untrained and Kvasir-trained (K) methods

Table 3 Mean Spearman agreement between clinical datasets by level (Dice), wound-pair agreement, and the modality difference (same-modality minus across-modality agreement) with its 95% method-bootstrap interval, exact relabelling p (the smallest attainable p is 0.00014) and NSD value. Method sets: U untrained, F from scratch, K Kvasir-trained, M19 modern; + denotes pooling

Methods	n	Within dataset	Within source	Within modality [a]	Across modality	Wound pairs	Modality difference (interval)	p	NSD difference
U	18	0.95	0.67	0.67	0.07	0.35	0.60 (0.35 to 0.83)	0.00029	0.42
F	14	0.91	0.66	0.60	0.14	0.16	0.47 (0.21 to 0.72)	0.0046	0.56
K	14	0.88	0.54	0.68	0.16	0.36	0.50 (0.24 to 0.73)	0.00058	0.68
M19	19	0.92	0.55	0.28	0.10	0.21	0.22 (0.05 to 0.46)	0.0061	0.36
U+F	32	0.70	0.85	0.74	−0.08	0.52	0.83 (0.64 to 1.00)	0.00014	0.71
U+K	32	0.95	0.58	0.70	0.00	0.27	0.68 (0.58 to 0.75)	0.00014	0.72
U+M19	37	0.98	0.86	0.81	−0.28	0.82	1.10 (0.98 to 1.20)	0.00014	1.01

[a] Different sources only.

Modality structure is clear within the untrained and Kvasir-trained groups

Within the untrained and Kvasir-trained groups, datasets of the same modality agree more than datasets of different modalities: the modality difference is 0.60 (p = 0.00029) and 0.50 (p = 0.00058) (Table 3). Both groups are above the floor on most non-wound datasets. The from-scratch group shows a similar difference (0.47, p = 0.0046), but its group mean Dice is below 0.1 on six of the seven non-wound datasets, so, as for the modern group, its rankings there order near-zero scores and we do not count it as evidence. For the untrained and Kvasir-trained groups the difference holds without the three within-source pairs (0.52 to 0.60, p at most 0.00014), under NSD (0.42 to 0.68, p at most 0.00043) and with empty masks excluded (0.50 to 0.60, p at most 0.00058; Online Resource 1, Tables S2b, S2e and S2f). Colonoscopy datasets from three sources agree at 0.86 and 0.83 within these two groups, whereas the wound datasets agree at 0.35 and 0.36.

For the modern models the same statistic cannot be interpreted. Their group mean Dice is at most 0.07 on each of the seven non-wound datasets, so their rankings there order near-zero scores. The modality difference among the 19 modern models is 0.22 (p = 0.0061), but it varied with the training seed: for the ten original architectures with nnU-Net it is 0.47 with the seed-0 models, 0.02 with the seed-1 models and 0.18 with the seed-2 models (nine architectures), and 0.05 for the eight new architectures (exploratory; exact p 0.00087, 0.38, 0.022 and 0.33, uncorrected). We therefore do not count the modern group as evidence of modality structure.

Synthetic benchmark validity follows the method pool

Generator B at texture amplitude 40 agrees with the four wound datasets at 0.65 (0.44 to 0.80) over the pooled untrained and modern methods, a value that would usually be read as good validation (Fig. 2, Table 4). Among the modern models alone it agrees at 0.21 (−0.17 to 0.48), and the wound datasets agree with each other at 0.21. The difference is 0.00 (−0.41 to 0.30; hierarchical −0.36 to 0.27), an interval too wide to say whether the benchmark is a better or a worse proxy than another wound dataset. The benchmark itself ranks the modern models reproducibly (mean agreement between its three synthetic seeds 0.93), so its low agreement with the wound datasets is not due to its 45-image size. Over the pooled set the benchmark agrees with each wound dataset less than the other wound datasets do (difference −0.17

on average, −0.38 to −0.02; methods resampled only, because untrained methods have no per-image synthetic scores). Among untrained methods it agrees at 0.53 (0.22 to 0.74); for the from-scratch and Kvasir-trained groups the intervals include zero.

Validity per reference varies, although single per-reference values are imprecise. Among Kvasir-trained methods the benchmark agrees with AZH at 0.56 and with FUSeg at −0.34 (difference 0.90, 0.16 to 1.50). Among untrained methods it agrees with CO2Wounds-V2 at 0.80 and with AZH at 0.32 (difference 0.49, −0.03 to 1.05). Among modern models its agreement with single wound datasets ranges from 0.05 to 0.47 (Table 4). Reported against a single dataset, the same benchmark and pool could be described as valid or invalid.

Background texture raised validity only in pooled sets, and not in all of them. Moving generator B from amplitude 11 to 40 raises mean benchmark agreement by 0.33 (0.09 to 0.59) in U+M19, with every per-reference interval above zero, and by 0.21 (0.08 to 0.36) in U+F, but not in U+K (0.05, −0.04 to 0.14). Within the groups the change is 0.08, 0.03, −0.03 and −0.08 (U, F, K and M19), and only 1 of the 16 per-reference intervals excludes zero; with no effect about 0.8 would be expected, although the 16 intervals are not independent.

Benchmark agreement among modern models also varied with the training seed. With single-seed versions of the original architectures (ten models; nine at seed 2) it ranges from 0.23 to 0.40 over the three seeds, for the new architectures (eight models; seven at seeds 0 and 2) from −0.15 to 0.19, and for the full single-seed modern group from 0.20 to 0.25; the agreement between the wound datasets themselves moves with the seed as well (0.05 to 0.28 for the original architectures). Leaving out any one modern method gives benchmark agreement between 0.12 and 0.32. Comparisons between generator families also changed with the pool and the references (Online Resource 1, Table S5f).

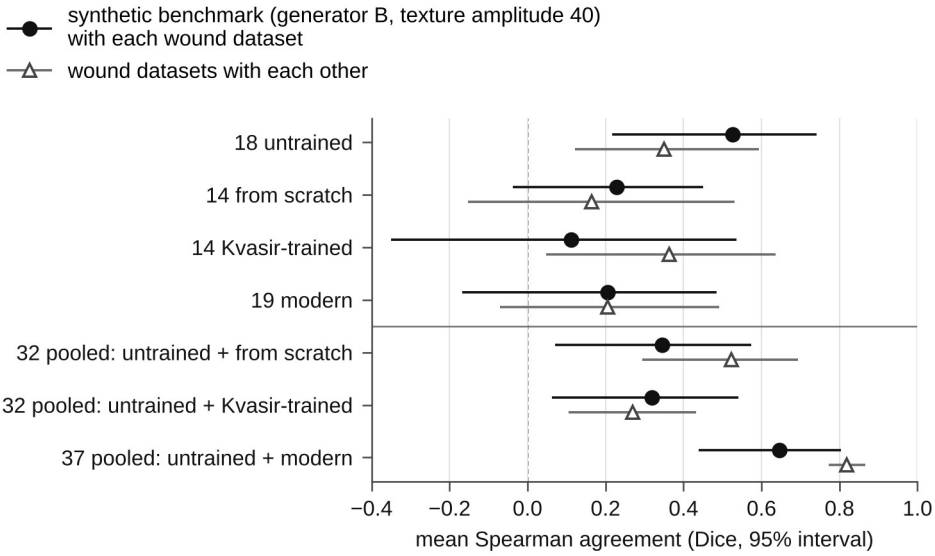


Fig. 2 Agreement of generator B (texture amplitude 40, mean over three synthetic seeds) with the four clinical wound datasets, and agreement of those datasets with each other, for each group and pooled set (Dice; 95% method-bootstrap intervals)

Table 4 Agreement of generator B (amplitude 40, mean over three synthetic seeds) with each wound dataset by method set, the mean over the four, the mean agreement of the wound datasets with each other, and the difference between the two (Dice; 95% method-bootstrap intervals)

Methods	Medetec	CO2Wounds-V2	FUSeg	AZH	Mean (interval)	Wound with wound	Difference (interval)
U	0.59	0.80	0.39	0.32	0.53 (0.22 to 0.74)	0.35	0.18 (−0.07 to 0.32)
F	0.05	0.47	0.10	0.30	0.23 (−0.04 to 0.45)	0.16	0.06 (−0.42 to 0.41)
K	0.16	0.06	−0.34	0.56	0.11 (−0.35 to 0.54)	0.36	−0.25 (−0.65 to 0.08)
M19	0.07	0.47	0.05	0.23	0.21 (−0.17 to 0.48)	0.21	0.00 (−0.41 to 0.30)
U+F	0.34	0.63	0.21	0.21	0.35 (0.07 to 0.57)	0.52	−0.18 (−0.42 to 0.04)
U+K	0.44	0.16	−0.03	0.69	0.32 (0.06 to 0.54)	0.27	0.05 (−0.11 to 0.17)
U+M19	0.64	0.76	0.58	0.61	0.65 (0.44 to 0.80)	0.82	−0.17 (−0.38 to −0.02)

Rankings among modern models are weak and noisy

Among the 19 modern models, the four wound datasets agree at 0.21 on average (−0.07 to 0.49; hierarchical −0.10 to 0.48), and the AZH control agrees with AZH at 0.92 (hierarchical 0.46 to 0.97). Under NSD wound-pair agreement is 0.48 (0.17 to 0.73; hierarchical 0.12 to 0.70). The models are close together: on each wound dataset the 19 method means lie within a band 0.07 to 0.21 Dice wide, and the median gap between neighbours in the ranking is 0.001 to 0.006.

Training noise is large (Fig. 3a, Table 5). Two seeds of the same 16 architectures, scored on the same wound images, agree at 0.44 on average (hierarchical interval 0.19 to 0.65), from 0.26 to 0.59 on the individual datasets, against 0.83 on the control images. Retraining one architecture with another seed moves its dataset mean with a pooled standard deviation of 0.010 to 0.033 Dice, several times the median gap between neighbours. The pre-specified rule required same-dataset cross-seed agreement above 0.7. It was not met (0.44). Cross-seed agreement was also not as low as cross-dataset agreement (0.09), so neither branch of the rule applies, and the comparison of the two that follows was not specified in advance.

The datasets nevertheless disagree more than the seeds do. Agreement between different wound datasets at different seeds is 0.09 (hierarchical −0.12 to 0.33), so the gap to same-dataset agreement is 0.35 (0.11 to 0.57; hierarchical 0.11 to 0.61). The gap was 0.50 (hierarchical 0.08 to 0.81) for the nine original architectures and 0.08 (hierarchical −0.16 to 0.49) for the seven new ones. The difference between these two gaps, 0.42 (−0.07 to 0.83), is too uncertain to attribute the gap to one subgroup; the new architectures also agreed less across seeds on the same dataset (0.27 against 0.55). Part of the gap reflects the shared test images: on random halves of each dataset, same-dataset cross-seed agreement was 0.45 when both seeds were scored on the same half and 0.36 when they were scored on different halves. Under NSD the gap is 0.14 (−0.01 to 0.30). Wound-pair agreement was 0.17 for single seeds (mean over three) and 0.17 for the seed averages, a difference of 0.00 (−0.14 to 0.17), an interval too wide to tell whether averaging seeds helps. The mean Dice of a single-seed model has a reliability (ICC) of 0.27 to 0.77 on the wound datasets, and a three-seed average one of 0.52 to 0.91 (point values; Online Resource 1, Table S4j).

The level of agreement also depends on how thresholds are set (Fig. 3b). With each model at its own FUSeg-tuned threshold, the protocol used throughout, wound-pair agreement is 0.21. With one common threshold of 0.5 for all models it rises to 0.40 (0.14 to 0.63; change 0.20, 0.03 to 0.35), and the rise has an

interval above zero for every common threshold from 0.45 to 0.90, reaching 0.62 at 0.90. A common threshold lets the ranking follow how confident each model's outputs are: how well a model's tuned threshold predicts its rank on the wound datasets (mean Spearman correlation) is 0.27 under the main protocol and 0.55 at a common threshold of 0.5. With a per-dataset oracle threshold for every model, chosen on the evaluation images, agreement is 0.12 (−0.14 to 0.39), not distinguishable from the main protocol (change −0.09, −0.28 to 0.09). On Medetec and CO2Wounds-V2 most oracle thresholds sit at the lowest grid value (92.3% and 96.2% of the models), so this setting reduces calibration differences only in part.

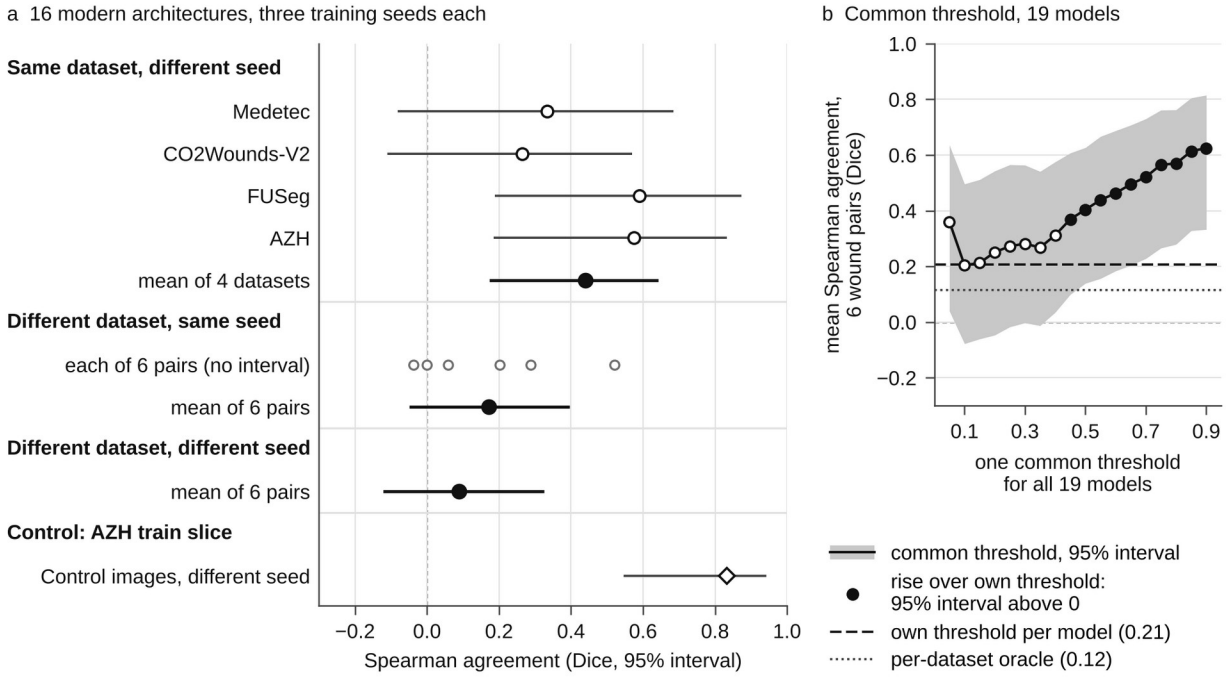


Fig. 3 Agreement among modern models on the four wound datasets (Dice). a: the 16 architectures with three training seeds; agreement between two seeds on the same dataset (per dataset and mean), between two datasets at the same seed (each of the six pairs, without interval, and their mean), between two datasets at different seeds, and between two seeds on the control images (AZH train slice), with 95% architecture-bootstrap intervals. b: mean wound-pair agreement of the 19 modern models when one common threshold is applied to every model, with 95% method-bootstrap band; filled markers show a rise over each model's own tuned threshold with interval above zero; the dashed line marks each model's own threshold and the dotted line a per-dataset oracle threshold

Consequences for choosing a modern model

The best model differs between wound datasets: MA-Net (ResNet-18) on Medetec, SegFormer (MiT-B2) on CO2Wounds-V2, MA-Net (ResNet-50) on FUSeg and SegFormer (MiT-B2) on AZH (seed averages), and with single-seed models the best architecture changes with the seed on three of the four datasets. Choosing on one wound dataset and deploying on another loses 0.048 Dice on average over the 12 ordered pairs (at most 0.119), against 0.063 for a random choice (Table 5); the paired difference is −0.016 (−0.033 to 0.003). For comparison, choosing on one random half of a wound dataset and deploying on the other half loses 0.001 to 0.024. With single trained models instead of seed averages the chooser did about as well as a random pick: 0.059, 0.077 and 0.053 against 0.060, 0.085 and 0.059 with the seed-0, seed-1 and seed-2 models.

Choosing on FUSeg, the test set from the clinic that supplied the training images, was the worst strategy: the model ranked first there lost 0.101 Dice on average on the other three wound datasets, including AZH from the same clinic, against 0.075 for a random choice on those datasets. Agreement between the datasets is consistent with this. Among the 19 modern models FUSeg agreed with Medetec at −0.12, with CO2Wounds-V2 at −0.13 and, from the same clinic, with AZH at 0.21, while Medetec and CO2Wounds-V2 agreed at 0.73 (Online Resource 1, Fig. S1). A good result on the internal test set therefore said little about performance elsewhere.

The synthetic benchmark picks nnU-Net at every texture amplitude, with a mean loss of 0.030. This pick depends on the training seed. With the seed-0 models the benchmark picks U-Net (EfficientNet-B3) (mean loss 0.067), an architecture whose synthetic Dice is 0.80, 0.67 and 0.47 at seeds 0, 1 and 2; with the seed-1 models the pick is nnU-Net again, but its mean loss is 0.051 because the best models change (Online Resource 1, Table S7b).

Mean Dice hides a failure mode that differs between the two training pipelines. On AZH the 18 pretrained architectures, trained and run on images letterboxed to 256 pixels, miss (Dice below 0.1) 28.7% (21.7% to 35.6%) of the 136 images with a non-empty reference on average, against 1.5% (0.0% to 3.7%) for nnU-Net, which works on native pixels; on FUSeg the rates are 2.6% and 4.6%. The new architectures missed fewer AZH wounds than the original ten (25.5% against 31.2%; difference 5.7 percentage points, 3.4 to 8.0 with these architectures fixed and −0.8 to 13.0 when architectures are also resampled). Even the best, SegFormer (MiT-B2), misses 12.3% (7.4% to 17.3%), and one architecture's AZH miss rate varied by up to 14.0 percentage points between its seeds; nnU-Net was trained once, so its interval does not include training noise.

AZH images are small wound crops pasted on a black canvas (85.6% of pixels black on average over the kept images, against 26.3% for FUSeg). The feature matches show that one AZH pixel covers 1.20 FUSeg pixels, so after letterboxing the pretrained models see AZH tissue at about 1.9 times the magnification of their training images, while nnU-Net, on native pixels, sees it at about 0.83 times. When the pretrained models do detect an AZH wound their mean Dice is 0.79, against 0.68 for nnU-Net. The share of the 51 pretrained models that miss an image correlates with its black-pixel share (Spearman 0.73) and with its lesion fraction (−0.46). Within each lesion-fraction tercile, from the smallest to the largest lesions, images with more padding than the median are missed by a larger share of models (0.78 against 0.23, 0.38 against 0.06 and 0.19 against 0.10). In 39 AZH images most pretrained models miss the wound and nnU-Net detects it; the reverse occurs in 0. These associations point to the difference in input scale between the two pipelines rather than to the architectures; they are correlational, because neither scale nor padding was manipulated.

Mean Dice and misses can point to different models. On AZH, SegFormer (MiT-B2) has the highest mean Dice (0.71 against 0.63 for nnU-Net) but misses 12.3% of wounds, against 1.5% for nnU-Net. Which model is better depends on whether a missed wound or a less exact outline costs more in the clinical task.

Table 5 Modern models: ranking reliability across training seeds, model selection and misses. Agreement is mean Spearman correlation over the four wound datasets (Dice). The interval type is given in each row: hierarchical (architectures and units resampled), architecture (architectures resampled), unit (units resampled with the models or candidates fixed). Losses are mean Dice below the best candidate on the deployment dataset

Quantity	Estimate	95% interval or detail
Ranking reliability (16 architectures with three seeds)		
Same dataset, different seed	0.44	0.19 to 0.65 (hierarchical)
Different dataset, different seed	0.09	−0.12 to 0.33 (hierarchical)
Gap (same dataset minus different dataset)	0.35	0.11 to 0.61 (hierarchical)
Gap, 9 original architectures	0.50	0.08 to 0.81 (hierarchical)
Gap, 7 new architectures	0.08	−0.16 to 0.49 (hierarchical)
Gap, original minus new architectures	0.42	−0.07 to 0.83 (architecture)
Different dataset, same seed (mean over the three seeds)	0.17	−0.05 to 0.40 (hierarchical)
Different dataset, seed-averaged models	0.17	−0.15 to 0.52 (hierarchical)
Model selection (Dice lost against the best candidate)		
Chosen on another wound dataset, 19 seed-averaged candidates (mean of 12 ordered pairs)	0.048	0.032 to 0.073 (unit); range 0.000 to 0.119
Random choice, 19 seed-averaged candidates (expected)	0.063	0.055 to 0.081 (unit)
Chosen on FUSeg, deployed on the other three	0.101	random choice 0.075
Chosen on another wound dataset, seed-0 models (18 candidates)	0.059	random choice 0.060
Chosen on another wound dataset, seed-1 models (19 candidates)	0.077	random choice 0.085
Chosen on another wound dataset, seed-2 models (17 candidates)	0.053	random choice 0.059
Synthetic pick, seed averages	0.030	nnU-Net
Synthetic pick, seed-0 models	0.067	U-Net (EfficientNet-B3)
Synthetic pick, seed-1 models	0.051	nnU-Net
Synthetic pick, seed-2 models	0.024	nnU-Net
Misses on AZH (non-empty reference, Dice below 0.1)		
18 pretrained architectures	28.7%	21.7% to 35.6% (unit)
10 original architectures	31.2%	23.9% to 38.3% (unit)
8 new architectures	25.5%	18.9% to 32.0% (unit)
Best architecture, SegFormer (MiT-B2)	12.3%	7.4% to 17.3% (unit)
nnU-Net	1.5%	0.0% to 3.7% (unit)

Robustness of the findings

The pooled-set findings survive the analysis choices (Online Resource 1, Table S6g). In all 12 conditions (three post-processing pipelines under Dice and NSD, four further NSD tolerances, and FUSeg at 256 pixels under Dice and NSD), U+M19 wound-pair agreement is 0.82 to 0.89 and exceeds the higher

within-group value by 0.23 to 0.47, each time with an interval above zero; agreement between wound and other datasets is −0.58 to −0.43; and stratified agreement (0.28 to 0.58) equals the size-weighted within-group average in every condition. The weaker U+F effect does not reach this standard in any condition (excess over the higher within-group value −0.16 to 0.17). The modality difference of every pooled set stays above zero in all conditions (U+M19 1.00 to 1.13).

Among modern models, wound-pair agreement stays within 0.12 to 0.26 under Dice (post-processing pipelines, resampled FUSeg and oracle threshold) and 0.31 to 0.53 under NSD (post-processing pipelines, tolerances and resampled FUSeg). Apart from the common threshold described above, no change has an interval that excludes zero; the largest were the oracle threshold under Dice (−0.09, −0.28 to 0.09) and the 5-pixel tolerance under NSD (−0.17, −0.42 to 0.09), so changes of that size cannot be ruled out. Post-processing barely reorders the modern models under Dice (lowest rank correlation between pipelines on a dataset above the floor 0.90), more under NSD (0.73). Excluding empty reference masks leaves the modality differences nearly unchanged (U+M19 1.10 against 1.10, M19 0.23 against 0.22) and wound-pair agreement nearly so (U+M19 0.81, M19 0.24).

Discussion

What a validity figure measures

A rank correlation between a synthetic benchmark and clinical data is usually read as a property of the benchmark. In our data the same benchmark scores 0.65 or 0.21 depending on whether classical methods are in the pool, and 0.80 or 0.32 among untrained methods depending on which wound dataset is the reference. The high pooled value comes from the gap between groups. A benchmark on which modern models beat classical ones, as they do on every wound dataset, will correlate with any of those datasets over a pooled set, whether or not it can order the modern models among themselves. The stratification identity makes this exact: once the difference in level between groups is removed, what remains is the average of the within-group agreements, and among modern models that average is low.

Ordering modern models among themselves is the question a vendor, a hospital or a regulator usually faces. Among modern models the wound datasets agree with each other at only 0.21, and the benchmark agrees with them at 0.21. The interval of the difference runs from −0.41 to 0.30, so our data cannot tell whether the benchmark is a better or a worse proxy than another clinical dataset. When clinical datasets agree this weakly, a validation study against them has little power to show that a benchmark is good, and a validity figure is informative only next to the agreement between clinical datasets for the same methods.

Modern models: dataset differences, training noise and thresholds

Three things limit agreement among modern models. First, the models are close: the median gap between neighbouring models is below a hundredth of Dice on each wound dataset, much less than retraining moves a model, as reported for other benchmarks [27]. Two seeds of the same architectures rank them only moderately alike on one dataset. Second, the datasets may differ. Over the 16 architectures two wound datasets disagree more than two seeds do (gap 0.35, hierarchical 0.11 to 0.61). The gap was larger for the original architectures than for the new ones, but the two subgroups cannot be told apart at these sizes; part of the gap reflects image sampling, because two seeds scored on the same images share it; and for the original architectures seed and session cannot be separated. The pre-specified rule was not met, so the evidence supports only this narrower statement. Third, the threshold protocol matters. A common threshold raises agreement, but the raised ranking follows how confident each model's outputs are [29], a trait that carries across datasets, rather than which model segments wounds best; an oracle threshold,

which reduces calibration differences, did not raise agreement. Our main protocol tunes each model's threshold on held-out images from its training domain, as is usual before deployment, and the low agreement applies to that protocol.

For practice, a single dataset, a single training seed and a single threshold protocol can each change which modern model looks best. In our data the best model differed between wound datasets and changed with the seed, the synthetic benchmark's choice changed with the seed, a model chosen on one wound dataset did only modestly better on another than a random choice, and with single trained models hardly better. Selecting on the test set from the training clinic was worse than a random choice, which is the situation of a hospital that accepts a vendor's internal test result. Mean Dice and the share of missed wounds also pointed to different models.

Relation to prior work

The composition effect is Simpson's paradox [22, 23], and its role in benchmark agreement is established for language models [18] and for robustness of image classifiers, where models trained on different data lie on different lines relating in-distribution and out-of-distribution accuracy [19–21]. Our contribution is to measure its size in medical segmentation; to show that it inflates agreement between datasets of the modality on which the groups differ most in level, here wound photography, as well as creating disagreement across modalities; to give the exact identity behind stratification; and to show the consequence for validating synthetic benchmarks. That training seeds and small test sets make model comparisons uncertain has been argued for machine learning benchmarks and for medical imaging [27, 28]; here the seed variation is large enough to change the validity figure of a benchmark and the model it selects. The task dependence of synthetic-to-real transfer reported for CT [17] and the strong transfer reported for programmable lesion trials [3] are consistent with our finding that validity is not a fixed property of a synthetic set. Reporting intervals over methods follows recommendations to quantify uncertainty in medical imaging evaluation [62, 63]. Finally, the advice to maximise variation in synthetic data comes from training, where domain randomisation helps models transfer to real images [64–66]. A benchmark has a different job: it should order methods as the deployment data would, which argues for matching the deployment distribution rather than exceeding it.

Why wound photography differs

Colonoscopy datasets from three centres agree strongly among the untrained and Kvasir-trained methods, while wound datasets do not. Colonoscopy frames share a device class and a field of view. Wound photographs have no standard framing: the median lesion fraction of the four wound datasets ranges from 0.007 to 0.098, two are different crops of one clinic's photographs, and one was taken on smartphones in a leprosy programme. The AZH crops are mostly black padding, and that framing is associated with whether the pretrained models detect the wound at all. Dice depends on lesion size [6, 67]. Under NSD, wound-pair agreement rises from 0.35 to 0.62 among untrained methods and from 0.16 to 0.31 among from-scratch methods, but falls from 0.36 to 0.26 among Kvasir-trained methods, so framing conventions may contribute to the disagreement [68, 69], but the evidence is mixed.

Recommendations

1. Report a benchmark's agreement within the group of methods it is meant to compare, and name that group.

2. Report agreement against several clinical datasets, as a matrix, together with the agreement between those clinical datasets and a within-dataset control.

3. Report intervals that resample methods and the images or photographs of each dataset.

4. For trained models, train several seeds and report how well the ranking on one dataset is reproduced across seeds before interpreting differences between datasets.

5. State the threshold protocol, because a common threshold can create agreement that reflects calibration rather than segmentation quality.

6. Report failure rates, such as the share of missed lesions, next to mean Dice.

7. Do not select a model on a test set from the training clinic alone; check the choice on at least one external dataset.

8. If a mixed pool cannot be avoided, report the within-group agreements directly; stratified agreement is their size-weighted average and adds nothing beyond them.

Limitations

The modern sets are small (19 methods; 16, 9 and 7 architectures in the seed analyses), so Spearman correlations move in coarse steps and intervals are wide. Several statements about modern models rest on intervals that include zero.

The pretrained models and nnU-Net differ in input resizing, augmentation, training length and threshold, so their comparison is between two pipelines, not between architectures.

In the second session 3 of the 5 MA-Net trainings diverged under mixed precision and were not imputed. MA-Net on ResNet-18 is averaged over two seeds and MA-Net on ResNet-50 is a single model, so these columns carry more training noise than the others, although they are the best seed-averaged models on Medetec and FUSeg.

Seed 0 of the ten original architectures was trained in a different GPU session from seeds 1 and 2, so for these architectures the effects of the session and of the seed cannot be separated. The first session did not record the PyTorch and timm versions.

nnU-Net was trained once, in its 2D configuration, for 100 epochs rather than the default 1,000, with its threshold fixed at 0.5. Its scores carry the noise of a single training, which matters because it was the synthetic benchmark's pick in most settings. The pretrained architectures were trained on 392 of the 810 FUSeg training images at 256 pixels; they represent current architectures, not every deployed system.

The level of agreement among modern models depends on the threshold protocol. Our statements about low agreement apply to each model at its own tuned threshold; a common threshold gives higher agreement, and the oracle threshold, chosen on the evaluation images and limited by the grid, does not remove calibration differences.

The modern models score near zero Dice on all seven non-wound datasets, so their colonoscopy agreement and modality difference rank floor-level scores, and the from-scratch group is at the floor on most of these datasets too. The untrained methods are at the floor on FUSeg (group mean 0.06), so their wound-pair agreement includes rankings of near-zero scores on that dataset; without FUSeg it is 0.35.

The synthetic benchmarks are parametric, contain 45 images each and were built at three synthetic seeds; the benchmark analyses rest on one generator family at one texture amplitude. Learned generators such as diffusion models were not tested. Benchmark intervals for sets that include untrained or from-scratch methods resample methods only.

A fifth wound dataset was excluded before scoring because many of its masks did not match their images. Wound photography is therefore represented by four datasets from three sources, two of which share one clinic, and statements about wound datasets rest on 5 pairs from different sources. Of the kept AZH test images, 57 are crops of FUSeg photographs that no model saw, so AZH and FUSeg are related beyond sharing a clinic.

Medetec's 152 images are 38 photographs with four variants each, so its effective sample is smaller than its image count; the AZH sets also contain several crops per photograph.

Images were not harmonised to one resolution across datasets, and NSD uses a fixed pixel tolerance, which is looser at lower resolution. Resampling FUSeg and varying the tolerance did not change the conclusions, but other datasets were not resampled. Untrained methods and probability-output methods receive different post-processing, which coincides with the main contrast between groups.

The seed, benchmark, selection and robustness analyses are exploratory and are not corrected for multiple comparisons. Intervals for selection losses resample images with the candidates fixed, and both losses are measured against the best observed model, which is biased upwards. The link between padding, scale and misses is correlational.

SAM and MedSAM received box prompts from the reference masks, an oracle that makes their scores incomparable with automatic segmentation; they were therefore kept out of every method set.

Conclusions

Agreement between clinical datasets about which segmentation method is better depends strongly on which methods are compared. Pooling classical and modern methods inflates agreement between wound datasets and turns agreement between wound and other datasets negative, and a synthetic benchmark validated over such a pool inherits the inflation. Among the modern models that practitioners choose between, wound datasets agree weakly, training noise is of the same order as the differences between models, and the threshold protocol shifts the level of agreement. When clinical datasets agree weakly with each other, a validation against them can say little about how well a synthetic benchmark orders the models. Validity should be reported within the group of methods to be compared, against several clinical datasets, with intervals over methods and images, and with the training seeds and threshold protocol stated.

Statements and Declarations

Ethics approval. Not required. The study used only publicly available, de-identified datasets distributed for research use.

Consent to participate and to publish. Not applicable.

Data availability. All clinical datasets are publicly available from their original distributors [30–39]. Online Resource 2 gives the per-method mean Dice and NSD on every image set, the agreement for every pair of image sets in every method set, and the values of the further analyses cited in the text. Online Resource 1 contains Tables S1 to S9 and Fig. S1. The synthetic generators, the benchmark images and the analysis code are available from the corresponding author on reasonable request.

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